



Marwadi
University

Marwadi University
Faculty of Technology
Department of Information and Communication Technology

**Subject: Digital Signal and
Image Processing
(01CT1513)**

Aim : Simulate discrete time sequences.

Open handed problem 1

Date:15-07-2026

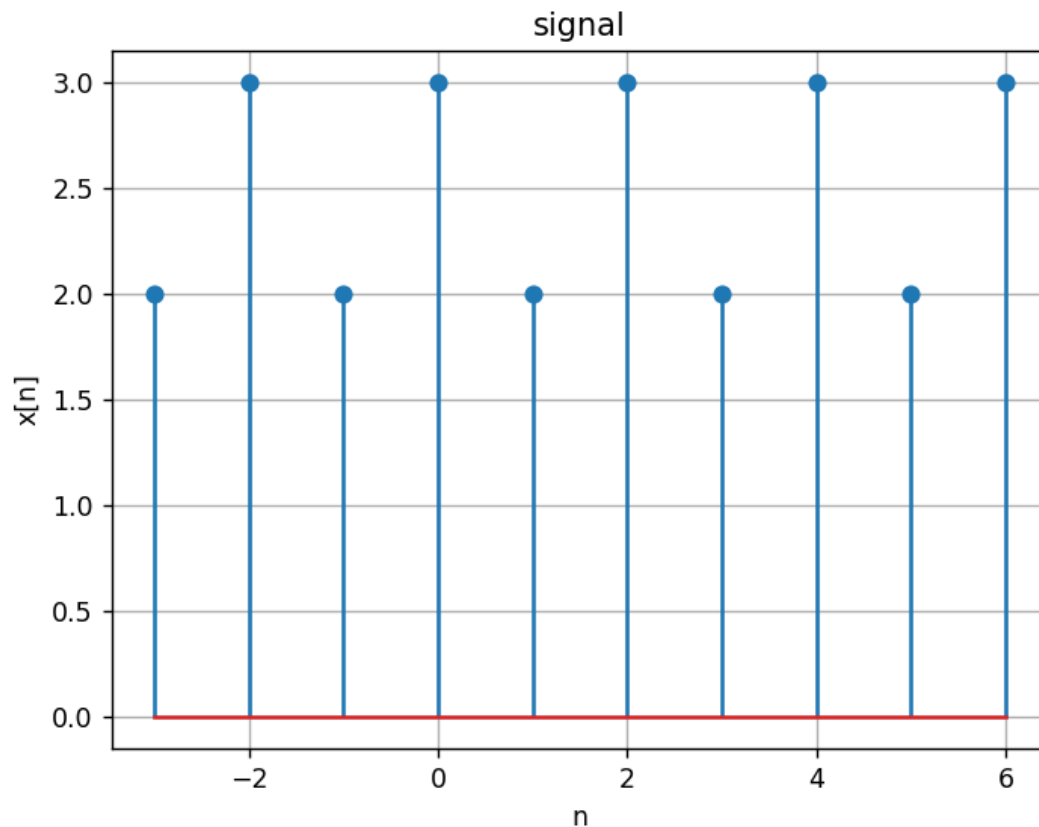
Enrolment No: 92400133131

```
Code 1 :- import numpy as np
import matplotlib.pyplot as plt

n = np.arange(-3, 7)
x = np.where(n % 2 == 0, 3, 2)

plt.stem(n, x)
plt.title("signal")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid()
plt.show()
```

Output :-



```
Code :- import numpy as np
```

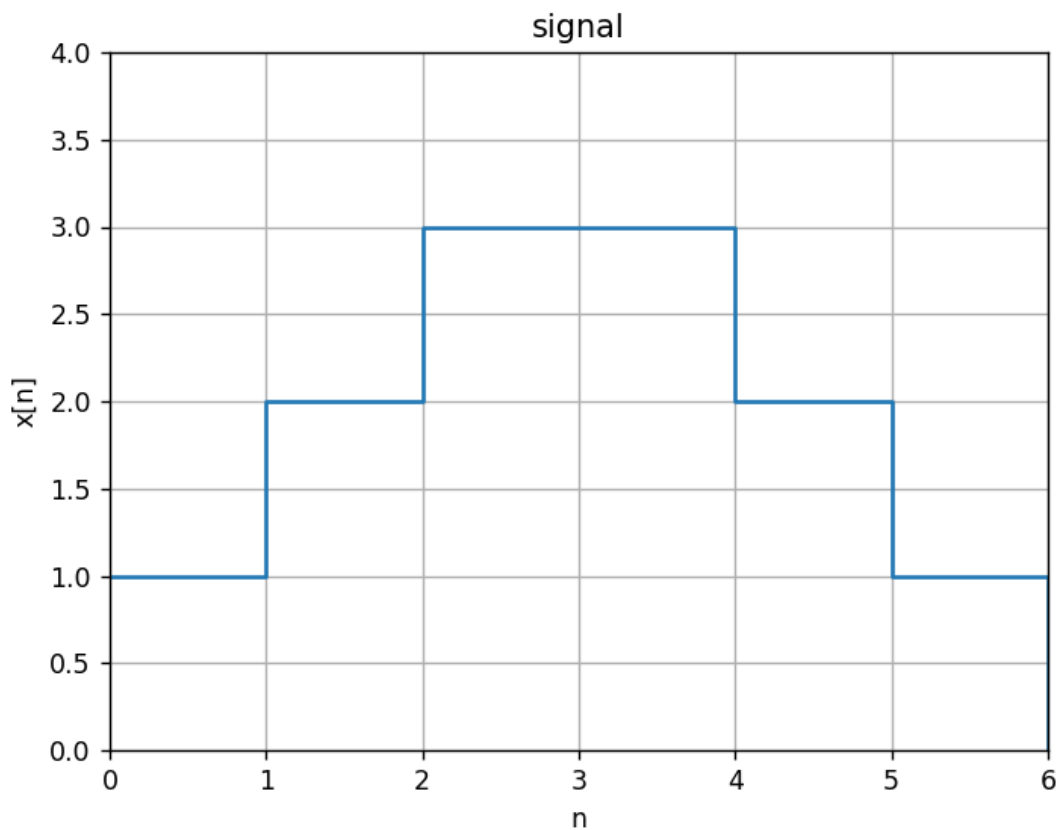
 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
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```
import matplotlib.pyplot as plt

n = np.arange(0, 7)
x = [1, 2, 3, 3, 2, 1, 0]

plt.step(n, x, where="post")
plt.xlim(0, 6)
plt.ylim(0, 4)
plt.title("signal")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid()
plt.show()
```

Output :-



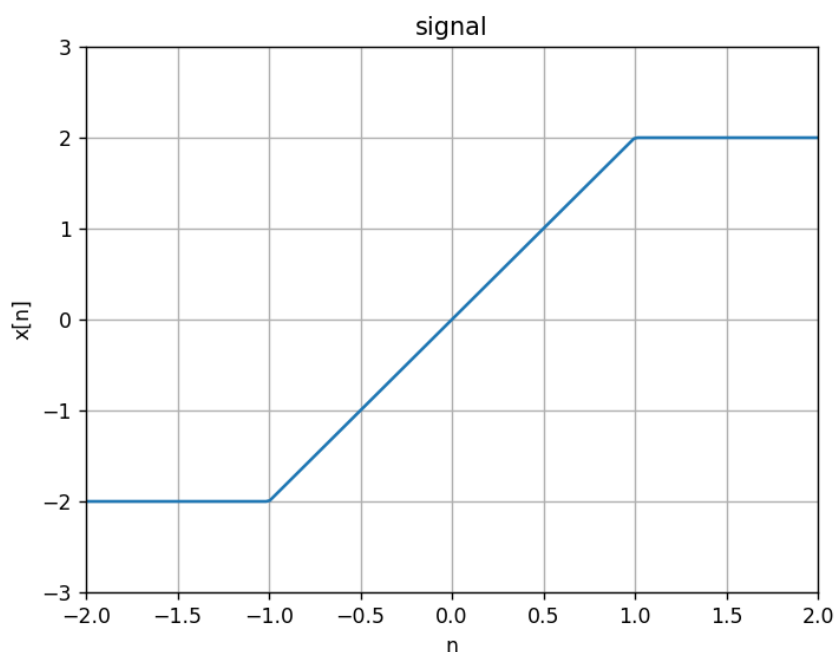
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```
Code 3:-import numpy as np
import matplotlib.pyplot as plt

n = np.linspace(-2, 2, 200)
x = np.pieceswise(n,
                  [n < -1, (-1 <= n) & (n <= 1), n > 1],
                  [-2, lambda n: 2*n, 2])

plt.plot(n, x)
plt.xlim(-2, 2)
plt.ylim(-3, 3)
plt.title("signal")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid()
plt.show()
```

Output :-



```
Code 4:-import numpy as np
```

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```
import matplotlib.pyplot as plt

n = np.arange(-2, 11)

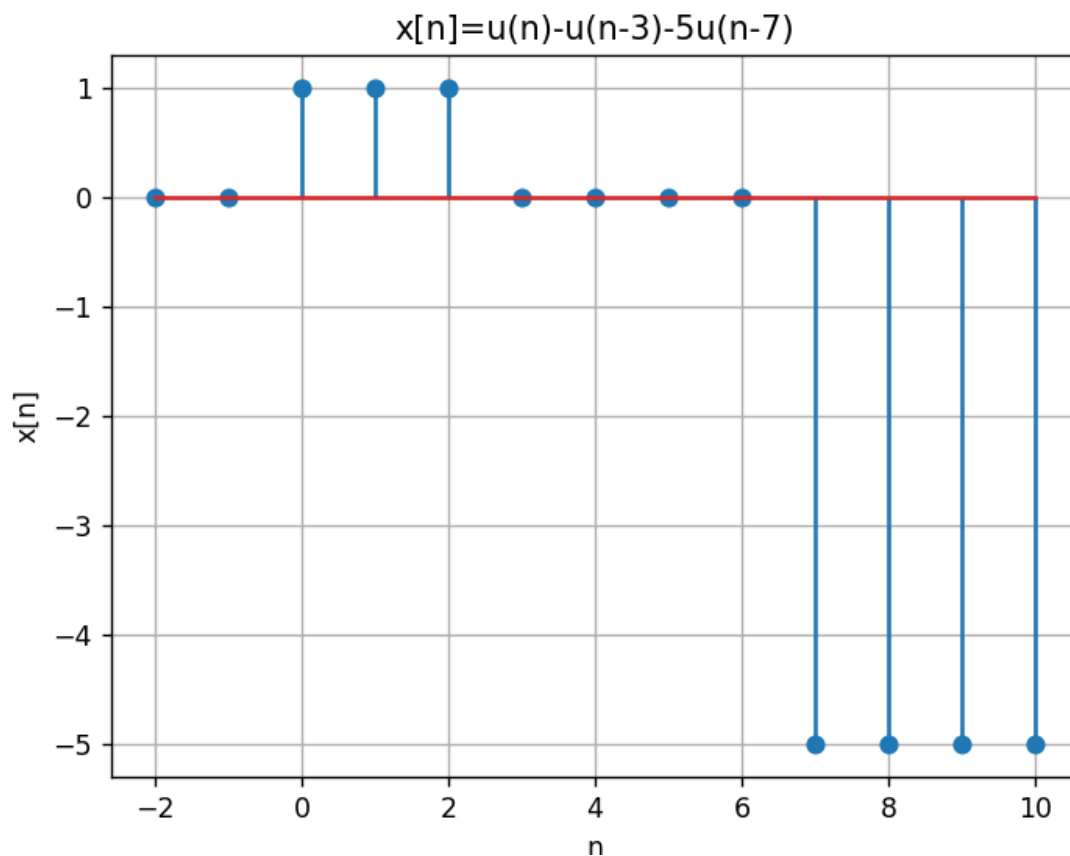
u = lambda x: (x >= 0).astype(int)

x = u(n) - u(n-3) - 5*u(n-7)

plt.stem(n, x)
plt.title("x[n]=u(n)-u(n-3)-5u(n-7)")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid()
plt.show()
```

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Output :-



```
Code 5:- import numpy as np
import matplotlib.pyplot as plt

n = np.arange(-3, 4)

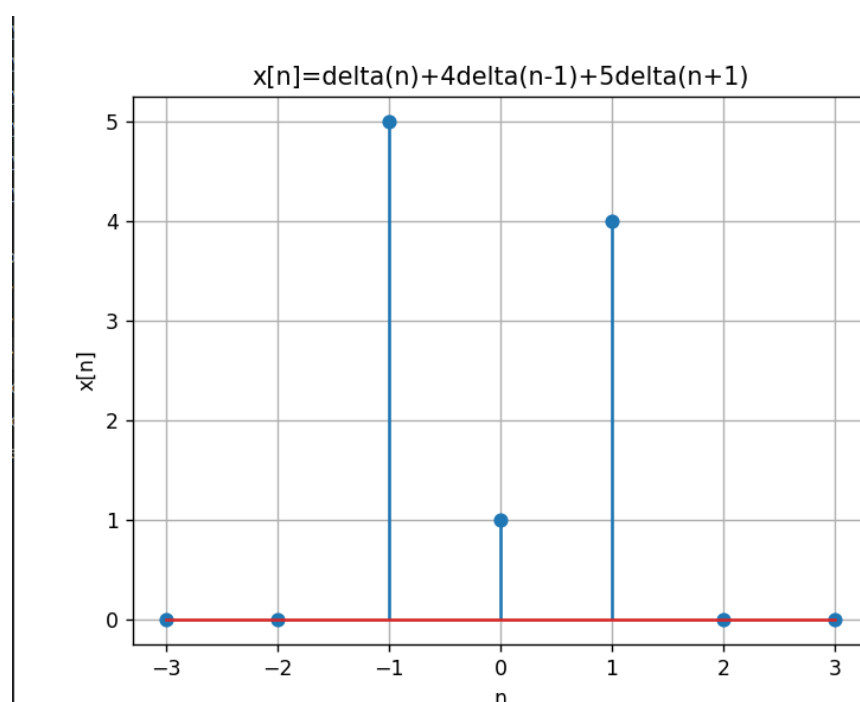
x = np.zeros(len(n))
x[n == 0] = 1
x[n == 1] = 4
x[n == -1] = 5

plt.stem(n, x)
plt.title("x[n]=delta(n)+4delta(n-1)+5delta(n+1)")
plt.xlabel("n")
plt.ylabel("x[n]")
```

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```
plt.grid()
plt.show()
```

Output :-



Github link :- <https://github.com/farhan-web404/Digital-Signal-and-Image-Processing/tree/main/lab%201>

Conclusion :- The simulation of discrete-time sequences helped in understanding how signals are represented and analyzed at discrete time intervals. Different types of sequences such as unit impulse, unit step, ramp, and exponential were successfully generated and observed. This experiment improved the understanding of basic discrete-time signal concepts and their behavior, which are fundamental in digital signal processing.